

A Covid-19 Tracker for Medical Front-liners

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Abstract—The management of Covid-19 affected patients is a very difficult task. The current healthcare system of India is not able to cope with the enormous flow of patients and is in a dire need for improvement. This implementation paper provides a system which will manage all the affected patients right from the time they are Covid-19 positive till the time they are treated and discharged. This paper includes all the technical details of a fully implemented healthcare management system which is a significant improvement in the current system. The proposed system is a cross platform multi user web app which can be used by multiple stakeholders to carry out smooth management of the patients. It consists of a lot of key features like dynamic location-wise patient status, an accurate tracking system of ambulances, a statistical trend analysis of patients and categorical report generation of patients. This system aims to help the medical Front-liners in efficient management of Covid-19 patients, and it is a common site for all the different health workers like field workers and medical officers to work together and fight against this deadly disease affecting our country.

Keywords—Covid-19, Geocoding, Analysis, Pandemic, Tracking, Ambulance, Maps, Clustering, Patients, Hospital.

I. INTRODUCTION

Since the beginning of the Covid-19 outbreak, healthcare professionals have truly been selfless warriors, giving everything to uphold their job responsibilities. The Covid-19 pandemic has added to the stress of an overburdened healthcare workforce. India has one medical doctor for every 1,404 people and 1.7 nurses per 1,000 people, according to the Ministry of Health and Family Welfare (MoHFW). The existing healthcare system for managing the patients is distributed, less efficient and slow. It is not capable of handling 90,000 patients daily and the 130 crores population including urban rural [1]. The Public Health Care system requires a reboot. It should be more centralized, data-driven and rapid. Field workers find it harder to attend to all patients directly. Even though the addresses of people in quarantine are given, their exact locations are not known. There is a need to develop a cross-platform application [2] which helps the government organizations, medical officers and field workers to digitalize and automate the entire process, analyze and derive insights

based on geographical clusters and locations. Hence, our project aims to create an integrated digital platform for the medical Front-liners which will help in assisting the governing bodies from the time a patient is found till they are discharged [3]. It will also provide daily, weekly and monthly insights.

II. METHODOLOGY

A. Background

Some of the existing systems are Aarogya Setu [4], Corona Warrior [5], CoBuddy-Covid19, COVID-19 Care Tamil Nadu[6], etc. We thoroughly analyzed the pros and cons of each of them. It helped us form the objectives of this project. Some of the lacunae were technical glitches, privacy, and use of native applications which have high release lag and requires respective expertise [7]. Also, there was very little use of the latest technologies..

Aarogya Setu uses Bluetooth, GPS, AI and data analytics and the prime goal is contact tracing. The salient features are self-assessment, user status, national news, E-pass registration and information dissemination. Corona Warrior had dedicated knowledge sections for different stakeholders and educational quizzes and games. The main aim was information sharing. CoBuddy-Covid19 had a location tracking module mainly for quarantined patients. It also included a face recognition module, but was abandoned due to privacy issues. Covid-19 Care Tamil Nadu consisted of a visual map, district-wise reports, containment zone indicator and government orders.

The major advantage of the above apps over our system was support of local languages and voice assisted app operation.

Our team visited the Thane Municipal Corporation war room to understand the problem and requirements better. We also checked the entire manual process currently being carried out for the Coronavirus patient tracking.

There were many features that we also liked from the existing systems, one of them being automated discharge report generation [8][9].

B. Proposed System

Our system aims at automating the process of all the stakeholders involved in the treatment of a Covid-19 patient. It is a standalone multi-user app and a website for the stakeholders which is cross platform and dynamic.

The proposed system consists of the following components: Android/iOS application, Website, Maintenance and support, Initial orientation and training, Routine changes and updates, and Database.

The system should be able to:

- Tag the geographical location and assign Health Centre
- Record the current condition of the patient and take relevant decisions on whether the patient needs to be hospitalized or not
- Check the availability of the required medical facility and allocation of the patient
- Monitor the process flow until the patient is given proper treatment[10]
- Perform analysis on geospatial data of the patients[7][11]

There are 6 stakeholders involved in this project. Following are the roles and responsibilities of each of them.

- War Room - Master User
- Medical Officer - Doctors who can prescribe bed type and assign field workers
- Field Worker - Go to the patient's place and fill the medical form
- Ambulance driver - Transport the patients
- Hospital - Admit and discharge the patients
- Police Officers - Take legal action on the non-contactable patients

The complete process can be viewed as below:

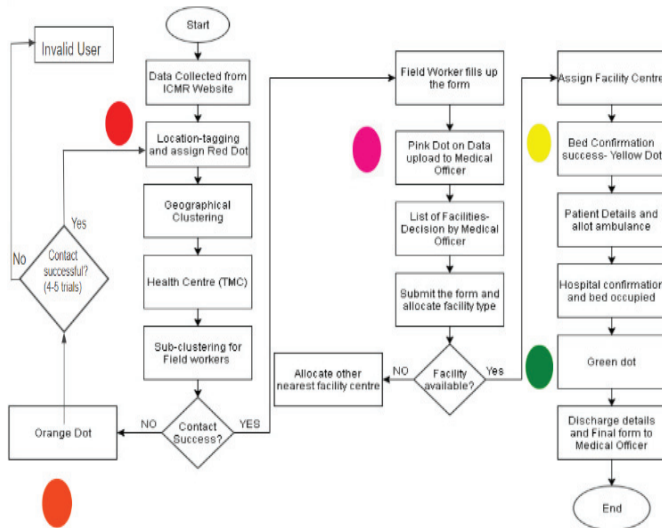


Fig. 1. Flowchart of the entire system

The process starts by getting patient's data from the ICMR website. It includes patient details who were tested Covid-19 positive the previous day. From the address field,

we extract the latitude and longitude of the patient [12]. Using these coordinates, we mark the precise location of the patient on the map with a dot whose color is initially red which indicates that the patient is a new patient. The governing body has certain groups of health centers each with their own area of jurisdiction. All the patients are assigned to a particular health center according to clustering algorithms.

After this, each field worker under a health center will be assigned a patient. The field worker then calls each patient assigned to them and confirms whether the address given by the patient is accurate and if the patient will be available in the given address at a specific time or not. If the patient does not answer or the address seems false, the particular patient details will be sent to the police who will help the health center in tracking down the patient. The patient's dot will then turn orange which signifies that the health center is unable to contact the patient and requires police assistance. If the patient has successfully confirmed their location, the field worker assigned to them will go to the particular patient's address. Upon reaching, the field worker will note down the patient's symptoms like SPO2, pulse and temperature, fill and upload the patient's form on the webapp. The patient's dot will now change to pink which signifies that the patient's form has been filled and uploaded for the medical officer to be reviewed.

After reviewing it, the medical officer will assign a facility type according to the patient's condition. The particular patient's facility request will be sent to the hospital assigned and they can either accept or reject a patient. If rejected, the medical officer will assign another nearest facility for the patient's treatment. If accepted, the patient will be assigned that hospital and the governing body will assign a particular ambulance according to the patient's condition. The patient's dot will now turn yellow which signifies confirmation of a bed for the patient. The ambulance driver will receive the patient information and the hospital information so that they can transport the patient to the concerned hospital quickly. The patient will also receive an SMS regarding the ambulance driver's details. When the patient reaches the assigned hospital, the patient's dot will change to green which signifies that the patient has been successfully admitted. The patient's treatment will now start and upon recovering, they will be discharged from the hospital.

III. MODULES

A. Data Collection and Preprocessing

The source of Data Collection is the ICMR (Indian Council for Medical Research) website. The data which is collected consists of Critical information of the Covid positive patients. The fields include Name, Address, Contact, Past medical History, Travel History, etc. Based on the data from the ICMR website is downloaded in the form of an excel document. Based on this data collected, the source data is collected and tabulated into the Database for further use. PHP is used to add the excel rows to the SQL database. Data cleaning process starts where null values are treated and imputed. The address format is corrected which is used for plotting the points on the map for Visualization. The patients with blank ICMR id, incorrect phone number, blank address are stored in a separate excel file so that the master user can

take action. Once pre-processing is completed, the data is ready for further use.

B. Geocoding and Geotagging patients

Google Maps help us to visualize the necessary data in one place, such as the patient's precise location, patient's name and some basic information along with the status of the patient. Visualization based maps enable the medical facilities to prepare in advance and can severely limit the impact of Covid-19[13].

To mark patients' location on the map, geographic coordinates are required. The Google's geocoding service converts text addresses into geographic coordinates i.e., latitude and longitude which later on can be used to place markers on a map or position the map. To distinguish between the current and the previous medical status of the patient, there is a need to assign dynamic markers as follows:

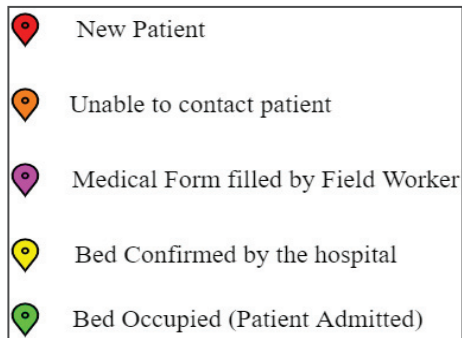


Fig. 2. Dot Color Specification

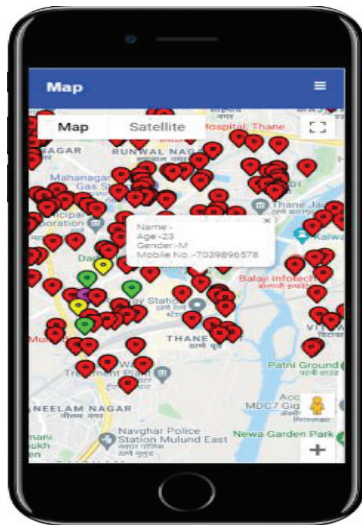


Fig. 3. Different Markers and Information Window

C. Assigning Health centers and field workers

The patient geolocation data is extracted from the database. Each government has divided their governing area into wards which we call Health-centers (HC) who take full responsibility of managing the patients under them. The health-center data is taken from the government in the form of .kmz files. The files were then converted to JavaScript arrays. Every patient Latitude longitude was checked against every HC co-ordinate array to determine its HC. The containsLocation() function of the Google maps API was used for this.

The field workers are manually assigned by the Medical officer using the webapp.

D. Hospital Management System

The Hospital Management system is controlled by every hospital by their respective admins. Hospitals can manage their bed counts and patient's acceptance/rejection along with patient discharge events [2]. There are different hospital facility types: CCC (Covid-19 Care Centre), DCHC (Dedicated Covid-19 HC), DCH (Dedicated Covid-19 Hospital), HC (Home quarantine), ICU, Oxygen, and Ventilator.

Hospitals can accept and reject new patients from their dashboard itself. This will make the entire process seamless. Upon accepting a patient the bed count in the hospital will be automatically updated. For the existing patients, the application will give hospitals the feature to discharge the patients. This will also update the bed count again. Once a patient has been discharged their entry will move to discharged patients for generating reports and further analysis [14].

E. Ambulance Management System

The Ambulance Management System is controlled by the Master User/Admin. As soon as the medical officer assigns a hospital to a patient, the Master User/Admin receives the request to allocate an ambulance to the patient. According to the needs of the patient, the Master User/Admin decides which type of ambulance should be allocated. The types of ambulances can be Cardiac, Ventilator, Oxygen, and Normal Ambulance. After selecting the type of ambulance, the Master User/Admin then selects the ambulance driver allocated to a particular patient.

For allocating the nearest ambulance from the patient's address our proposed system uses Google Map's Distance Matrix API. The nearest ambulance ID gets automatically mapped with the patient details using this API. The ambulance driver has its dashboard. Using this dashboard the ambulance driver can set this duty as ON/OFF. According to the duty preference of the ambulance driver the system helps the Master User/Admin in allocating an ambulance to the patient. After an ambulance is allocated to the patient, it is expected that the ambulance driver keeps himself online in the mobile app. This will help the Master User/Admin and the Medical Officer to get live tracking of the ambulance. Google Maps Direction API is used to find the fastest and most convenient route between the ambulance's location and the patient's address. A route with least congestion is selected.

For sending SMS to all the patients and ambulance drivers, our web app uses the D7 SMS API which accepts the contact details of the sender and sends a text message including the details respectively. All the information about the ambulance such as Ambulance number, Driver's Contact number, ambulance current location and estimated time to reach patient's location is sent to the patient and the medical officer via text message. Also, all the details of the patient such as the patient's contact number, patient's address and the hospital allocated to the patient is sent to the ambulance driver via text message.

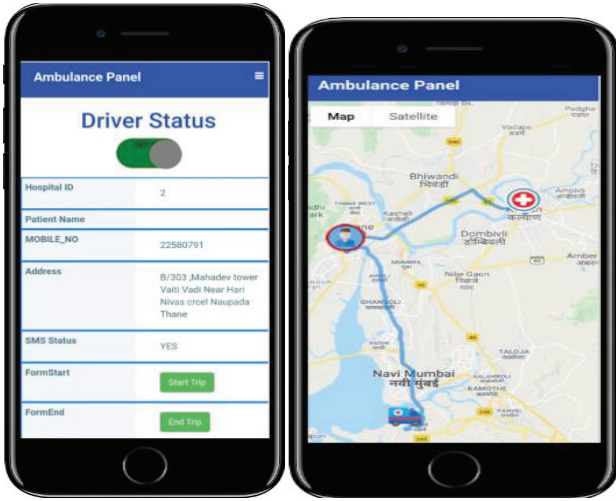


Fig. 4. Driver Dashboard and Routing

F. Report Generation

There were two types of automated reports generated for administration.

1) Patient discharge reports

After a patient is discharged, a discharge report in the form of pdf is generated. The PDFMake API is used to generate the report.

The project produces the pdf record on the fly from the MySQL database in this web application. A download interface will be given on the front-end or UI on which the end user will tap and will get an alternative to spare the created pdf record as per their chosen location. The system utilizes the PDFMake API for producing pdf reports. As we are not saving this pdf report anywhere within the physical area of the server, we are going to send the information within the Response object as an attachment.

Reports are generally created in PDF format since a PDF file could be a “read-only” document that cannot be modified without taking off an electronic impression while other designs like image, word, excel, etc. can effectively be modified without taking off an electronic impression [8].

Patient Discharge Report	
ICMR ID	7571
Patients Name	Mr. Tanmay Bhat
Age	35
Gender	M
Address	301, Diamond Wing, Lodha Belmonte, Laxmi Nagar Balkum Pada Thane, Thane, Maharashtra, 400607
Mobile No.	9876543210
Covid-19 Test	Positive
Date	13-11-2020
Health Center Name	Manpada
Field Worker	Mr. Akash Shukla
Form Filled Date	03-11-2020
Temperature	102 F
Pulse Rate	90
Facility Type	DCHC
Medical Officer	Dr. Naveed Kasturia
Hospital	Ruby Hospital

Fig. 5. Actual Discharge Report Generated

2) Custom Data report

The data can be downloaded in the form of excel using custom attributes according to the requirements from the webapp. Every health center needs to send an audit report to the higher authorities. The excel format is used to digitally send the documents. First the database rows are converted to octet stream binary file in RFC 2046 and then to excel. Content-Disposition with excel content-types to indicate we want to save rather than display.

The report can be generated of patients from a certain timeframe by selecting start date and end date. The categories like Health center allotted, Hospital allotted, and Ambulance allotted can be selected for filtering the rows of excel.

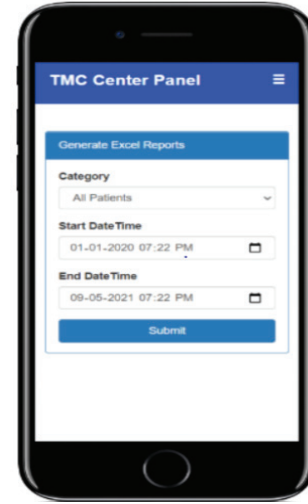


Fig. 6. Actual Custom Data Report Generated

G. Data Analytics

The analytics dashboard was uniquely curated for all users. Different metrics like age-wise patient distribution, gender-wise distribution are used. Patient’s age is an important attribute in determining the type of treatment. The comprehensive role-wise metrics can be seen in Table 1.

TABLE I. LIST OF METRICS

Master user	Health Center	Hospital
Gender-wise distribution	Contact status count	Bed availability
Patient dot colour-wise distribution	Form filled status	Insurance-wise distribution
Allocation of hospital and health-center status	Patient dot colour-wise distribution	Senior Citizen, Discharged Patients, Currently admitted, Critical Patients count
Health-center wise patient frequency	Gender-wise distribution	Ambulance Allocation status
Age-wise patient distribution	Hospital wise patient frequency	Age-wise patient distribution
Daily, weekly and monthly line charts for patient count	Age-wise patient distribution	Symptom severity distribution

The analysis data has the least access. For e.g. a health center can visualise data confining to only that health center.

Also, daily, weekly and monthly patient rate is displayed in the form of graphs. Google Charts library was used in JavaScript. The data was extracted from the SQL database in the form of an array. Then, different functions were applied to generate charts. The graphs give a nice view to the government to be ready for the future outbreak. The graphs were also displayed health center wise so the health center with the highest number of patients can be focused on. The co-morbidity aspect was also visualized to get an idea of what patients were more succumbed to due to Covid-19.



Fig. 7 Analytics on the dashboard of Master User

After a significant amount of data was collected, machine learning can be applied to predict the growth of Covid-19. Based on the symptoms recorded by the field worker, the severity of the disease can be predicted beforehand so that facilities can be arranged. Mortality and recovery can also be predicted. This becomes impactful in the cases of breathing equipment scarcity. Also, early diagnosis using Machine learning classification can be done [15].

IV. RESULTS

A. Improved Patient Tracking

With the help of above system, it will be easier and more accessible to track patients across the various stages of the events.

B. Efficient Hospital Management

Hospitals management will be easier as the complete process will be automated.

C. Faster Patient Allotment

One of the major concerns for positive patients these days is getting a bed. A digital solution will help patients get a comprehensive view of the type of beds available.

D. Efficient use of human resources

Though India has enough human resources, we are not using the power. Our system will help in efficiently assigning resources like police, field worker, etc. And make sure all are occupied.

E. Data centralization

The data stored in the database can serve as a central data pool. This data can then be mapped to any data warehouse for data mining and pattern prediction.

F. Emergency ambulance

The app can help in emergency allocation of ambulances in case a patient is critical.

V. APPLICATIONS

There are humongous real-world problems where this system can be used with little modification.

- Coronavirus vaccine dissemination
- Apart from pandemics like Covid-19 communicable diseases such as Dengue can be efficiently managed
- Vaccination drives for kids (Ex. Polio)
- Yearly medical checkup of senior citizens
- Emergency Accident patients
- National Screening of diseases like TB

VI. CONCLUSION

Thus, we created an efficient, robust, fully automated, data driven and centralized digital and integrated platform for the medical Front-liners. The powerful backend design and implementation along with the smooth and easy to use GUI of our application will help its stakeholders to control any situation effectively. Our project aims at integrating all the entities involved in controlling the spread of coronavirus.

In the future application of this system, we can couple it with interesting and informative visualization and analysis on historical data, thus enabling us to predict and provide better services and overcome more real world applications. This application can also be scaled to a national level and help people across India to control any kind of a pandemic situation. Patient interaction can be included in the system so that it reaches every type of stakeholder. Integration of a contact transmission module will serve more purpose to the existing system.

VII. FUTURE WORKS

Future work can be in the following systems.

- Pre-hospitalization diagnosis
- Appointment booking
- Adding telemedicine facility
- Adding diagnostics for easier clinical tests
- Better visualization on historical data
- Patient-interaction in the system
- Contact transmission module
- Mobile push notifications to the mobile app
- Multi-factor authentication for login
- Hashing the passwords using SHA for better security
- Local language support and voice user interface

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